

CS & IT ENGINEERING



Operating System

Memory Management

Lecture - 8

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Recap of Previous Lecture



Topic

Multilevel Paging

doubt ?



Topics to be Covered



Topic

Inverted Paging

Topic

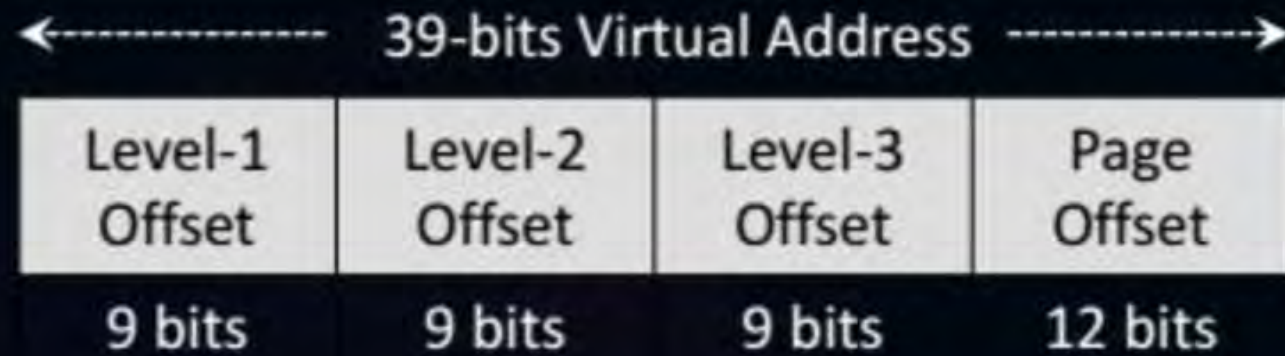
Hashed Page table



Topic : Question

Ans = 4108 KB

#Q. Consider a three-level page table to translate a 39-bit virtual address to a physical address as shown below:



L.A. = 31 bits

The page size is 4 KB = (1KB = 2^{10} bytes) and page table entry size at every level is 8 bytes. A process P is currently using 2 GB (1 GB = 2^{30} bytes) virtual memory which OS mapped to 2 GB of physical memory. The minimum amount of memory required for the page table of P across all levels is _____ KB across all levels?

← 31 →

p ₁	p ₂	p ₃	d
1	9	9	12

$$\begin{aligned}\text{outer} &= 1 \\ \text{middle} &= 2^1 = 2 \\ \text{inner} &= 2^1 * 2^9 = 1024 \\ \hline \text{Total} &= 1027\end{aligned}$$

$$\begin{aligned}\text{P.T. size} &= 1027 * 4KB \\ &= 4108KB\end{aligned}$$

single level paging



keep entire P.T.
in mm always



takes a lot of space mm

multilevel paging



keep required P.T. pages
in mm



saves a lot of space in mm



Topic : Question

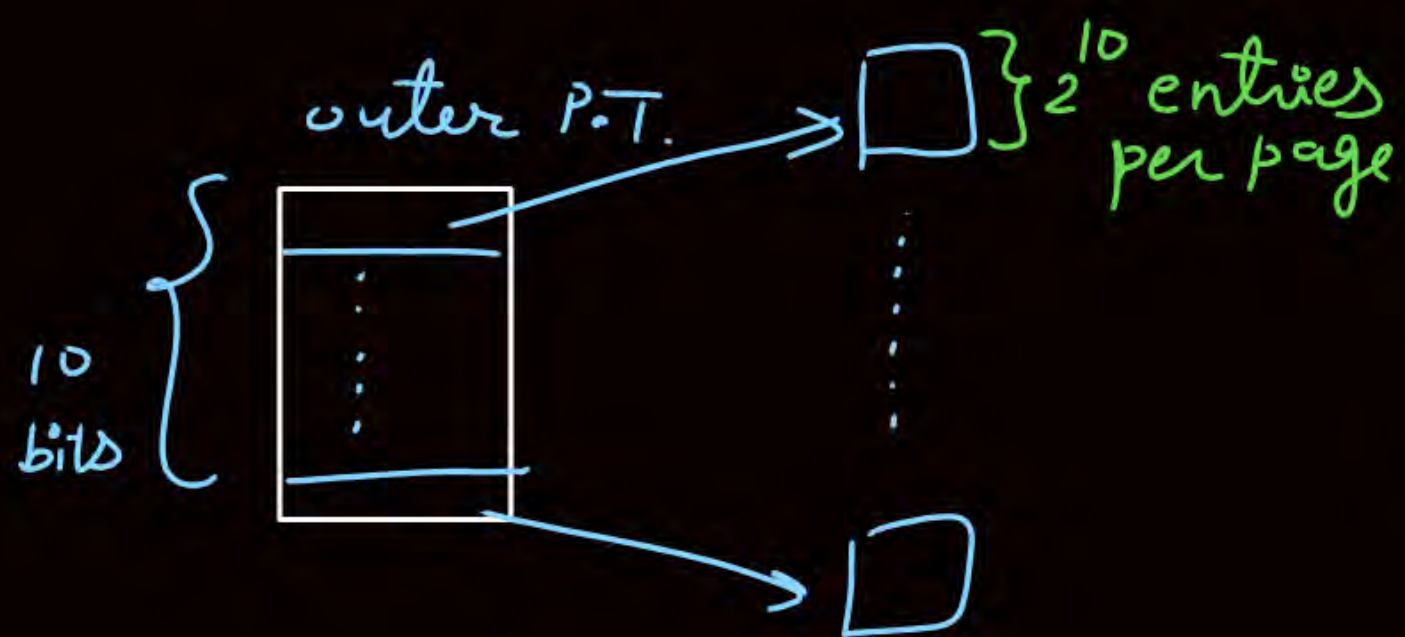
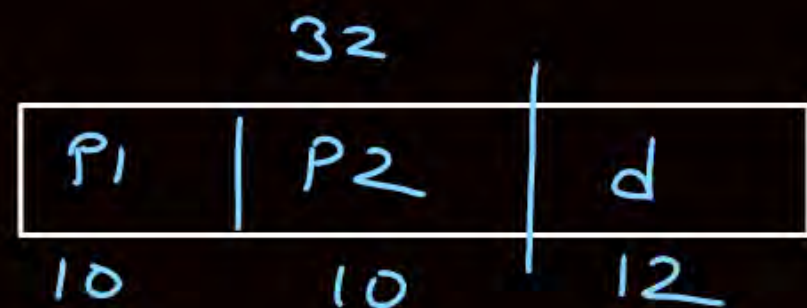
[GATE-2024]



#Q. Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume $1 \text{ KB} = 2^{10}$ bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X + Y$ is 1028?



Inner P.T.
has $2^{10} = 1024$
P.T. pages

min $\Rightarrow$ 1 outer P.T. page + 2 Inner P.T. pages to store 2000 P.T. entries

$$x = 3$$

max $\Rightarrow$ 1 outer P.T. page + 1024 inner P.T. pages

$$y = 1025$$

$$x + y = 1025 + 3 = 1028$$

Page table entry :-

outer P.T.

f	

Inner P.T.
page entries



frame no. + lesser
extra bits

Inner P.T.

f		...

f		

It might possible
that outer P.T.
entries are smaller
in size.

f	
⋮	

Actual process
P.T. entries

⇒ contain frame no. &
protection bits

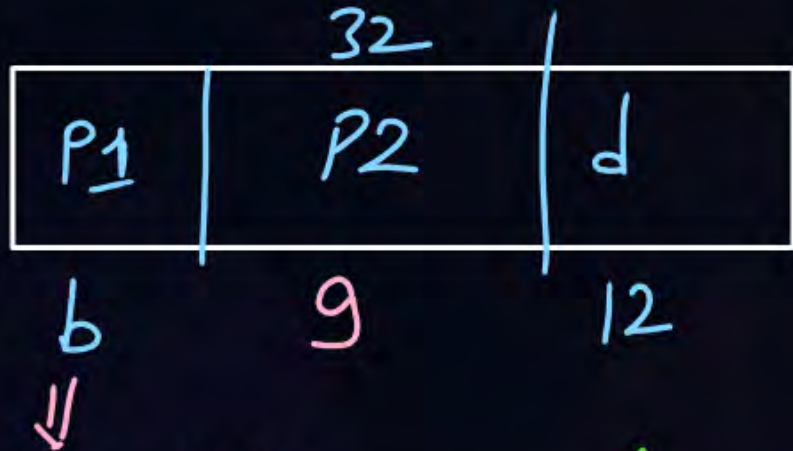


Topic : Question

LA = 32 bits [GATE-2025]



#Q. A computer system supports a logical address space of 2^{32} bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size. The value of b is 11?



$b = 11$ bits $\rightarrow$ outer P.T. entries must be smaller in size.

$$\begin{aligned} \text{no. of entries per page in inner P.T.} &= \frac{4096 \text{ B}}{8 \text{ B}} \\ &= 2^9 \end{aligned}$$

multilevel paging:-

2-level PT

$$E.M.A.T. = 3 * t_{mm}$$

2 times for P.T.
1 time for Content

for L-level Paging

$$EMAT = (L+1) t_{mm}$$



Topic : Inverted Page Table

Process

0	
1	
2	
3	
4	
5	
6	
7	

8 pages

Page Table

0	1
1	Invalid
2	0
3	2
4	Invalid
5	Invalid
6	3
7	Invalid

↑ indexing
⇒ page no.

↑ storing
frame no.

Inverted Page Table

0	2
1	0
2	3
3	6

↑ indexing
⇒ frame no.

↑ storing
page no.

4 frames
Physical Memory

0	Page 2
1	Page 0
2	Page 3
3	Page 6

no. of entries in inverted P.T. = no. of frames in mem

→ only one inverted P.T. needed for all process



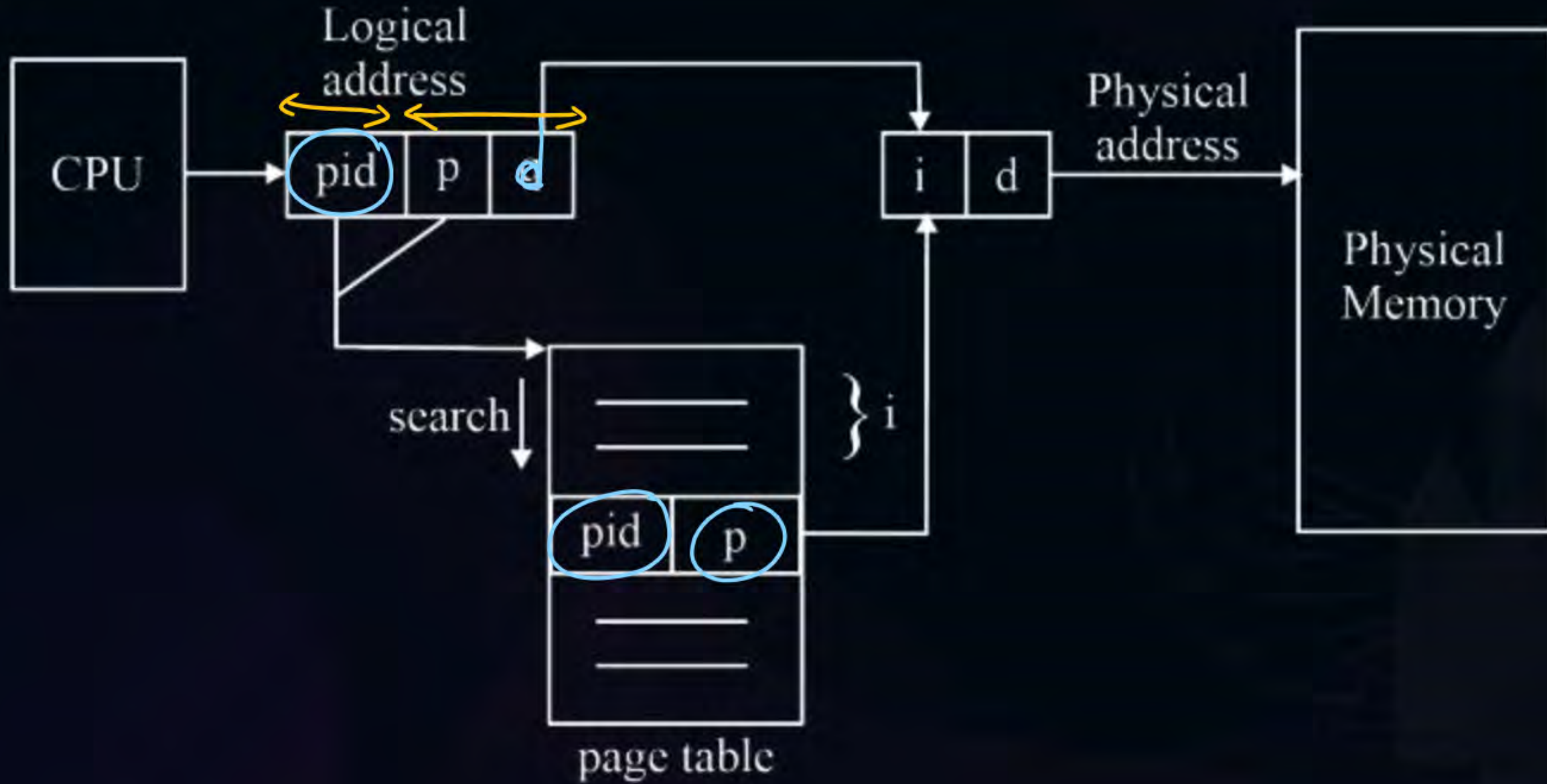
Topic : Inverted Page Table

Each entry in the page table contains the following fields:

1. Page number
2. Process id
3. Control bits (*protection*)
4. Chained pointer



Topic : Inverted Page Table

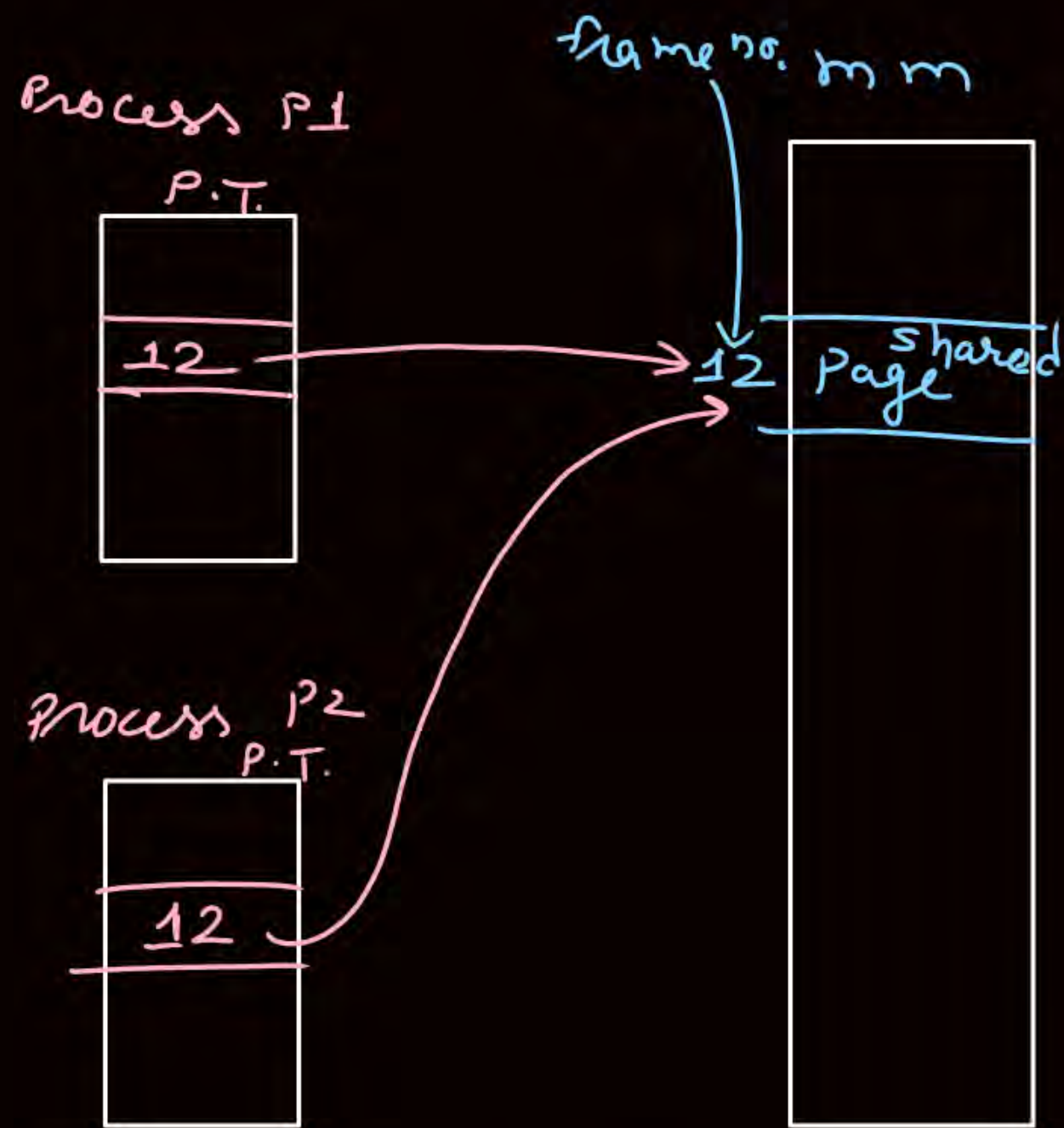




Topic : Inverted Page Table

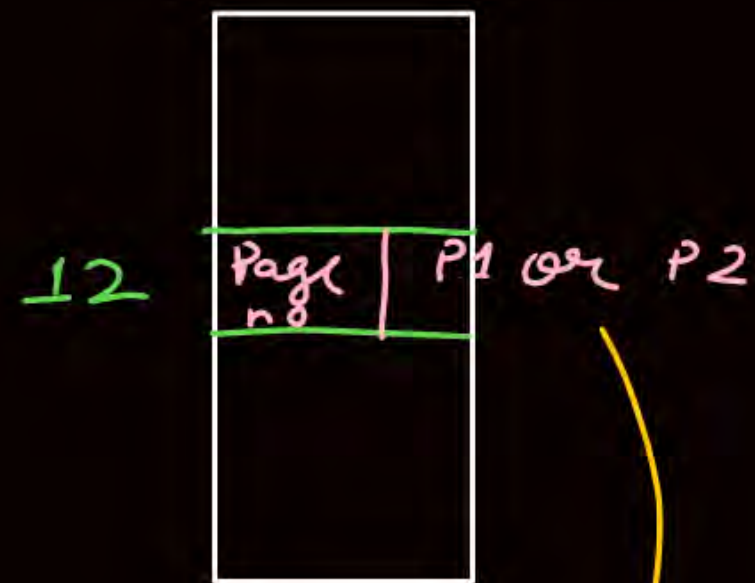
Advantages & Disadvantages:

1. Reduced memory space
2. Longer lookup time
3. Difficult shared memory implementation



Conventional P.T.

Inverted P.T.



so only one process
allowed to access the
Page.

example:-

Process size = 16 GB

mem size = 4 GB

Page size = 256 bytes

P.T. entry size = 4 B

$$\text{no. of pages in process} = \frac{16 \text{ GB}}{256 \text{ B}}$$

$$= \frac{2^{34}}{2^8}$$

$$= 2^{26}$$

$$\text{Conventional P.T. size} = 2^{26} * 4 \text{ B}$$

$$\text{Inverted P.T. size} = 2^{24} * 4 \text{ B}$$

no. of frames
in mem =

$$\frac{4 \text{ GB}}{256 \text{ B}}$$

$$= \frac{2^{32}}{2^8}$$

$$= 2^{24}$$



Topic : Hashed Page Table



In this virtual page, the number is hashed into a page table

This Page table mainly contains a chain of elements hashing to the same elements.

instead of having no. of indexes in P.T. for each page,
no. of indexes are reduces to counter invalid entries.

$$P.T. \text{ index} = H(\text{page no.})$$

ex:-
no. of pages in process = 8

conventional P.T.



↑
P.T. indexes

Hashed P.T.



Page no. P.T. entry

Page no. P.T. entry



Topic Hashed Page Table

Each element mainly consists of :

1. The virtual page number
2. The value of the mapped page frame.
3. A pointer to the next element in the linked list.

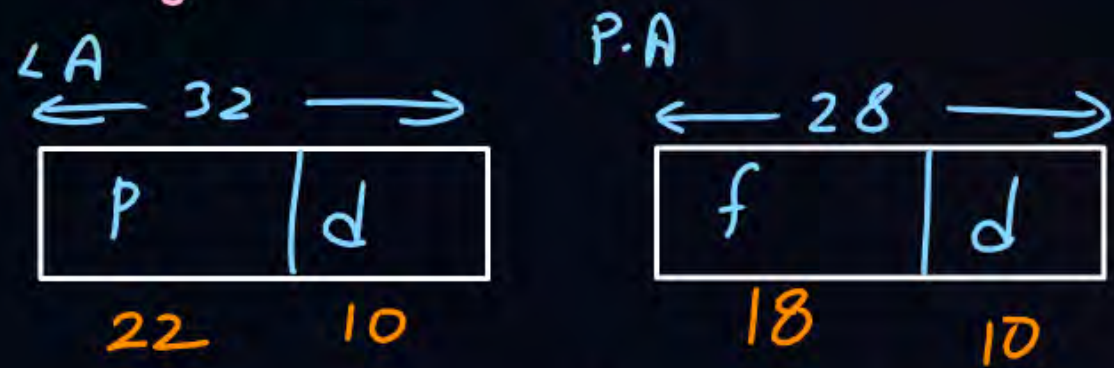


Topic : Page Size vs Page Table Size

→ As page size increases, P.T. size decreases

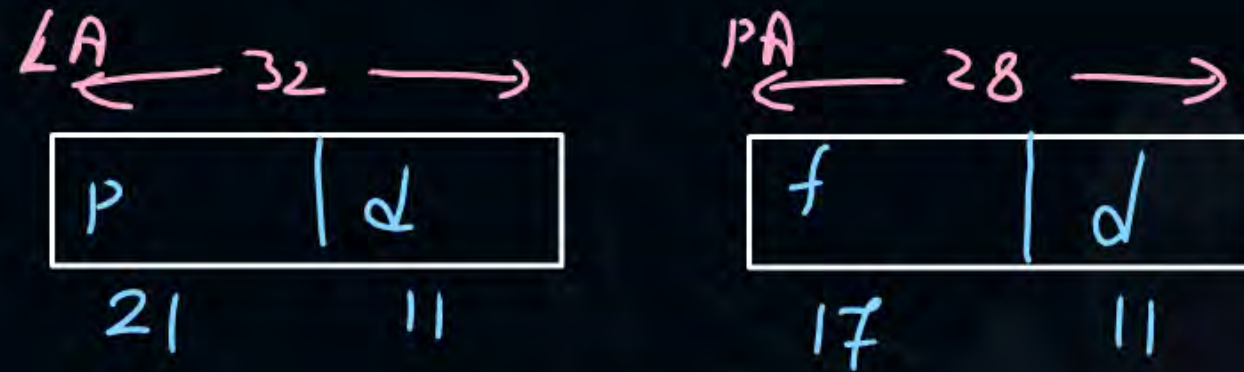
ex:-

Page size = 1KB



P.T. size = $2^{22} * 18$ bits

Page size = 2KB



P.T. size = $2^{21} * 17$ bits



2 mins Summary

Topic

Inverted Paging

Topic

Hashed Page table





Happy Learning

THANK - YOU

